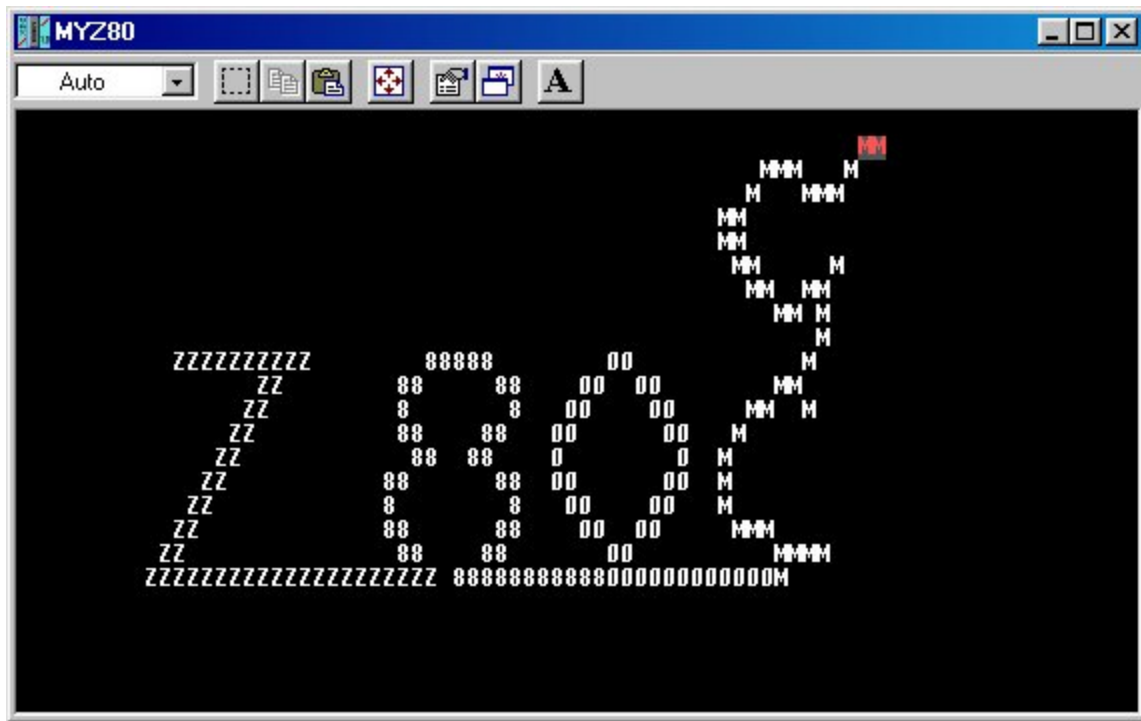




Mouse, the Language

[URL: http://primepuzzle.com/mouse/mouse3.html](http://primepuzzle.com/mouse/mouse3.html)

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Here's a program in Mouse which clears the screen (on a particular terminal ...)

```
~ cls.mse - Clears screen using a macro
#C;
$C 'Z '@ - !' @
```

1. Mouse programs end with a \$, but there may be things after the \$. These are called "macro definitions." Macros are like subroutines, procedures, functions etc. They are named using single capital letters, A thru Z. If this program had two macros, there would be another \$ followed by another, different, capital letter.

2. You invoke (ie. call) macros by using the pound sign (#) followed by the macro's letter. The semicolon (;) is the end-of-macro-call character. Macros may pass parameters (this example does not use parameters). A macro that did use parameters would be invoked by adding a comma (,) followed by the parameter for every parameter it supported. For example, #G,Q.,2; would be a call to macro G, passing it Q. and 2. The definition of the macro G might start like

```
~ find a prime factor of 1% using 2% as the initial candidate
```

```
$G 1%n:2%f:
```

The 1% would be replaced with the value Q. and the 2% would be replaced with 2.

We will have an example of this more complex kind of macro later in the tutorial.

All our simple macro C does here is push the character Z on Stack, then the character @ and then subtract @ from Z. The ASCII code for Z is 90 and the ASCII code for @ is 64. $90 - 64 = 26$. This (non-printing) "character" is then sent to the screen. Sometimes named ^Z, this "character" is what clears the screen for this terminal.

Macro definitions end with the @ character (not to be confused with the character @ which happens to be part of the definition of our particular macro!)

So, what's with that Z80 Mouse up top?! We may be jumping ahead a bit here, but ...

1. It illustrates the use of macros that position the cursor, set the foreground color etc. for the "MyZ80 terminal" in the version 1.21 MyZ80 emulator. The program works in the version 1.11 emulator but some of the fancier calls (involving color for instance) do nothing. Please visit <http://primepuzzle.com/mouse/hints1.htm> to learn about "MaxZ80," a full-up "Z-System" which uses the version 1.21 emulator.

2. It introduces the concept of "global" variables.

Most Mouse interpreters implement the concept of "global" variables. The one we use here does, but you should be aware that if you are running your program under one that does not, it won't work! A "global" variable is one that is known throughout the program. Upper case letters, A thru Z, are used for these variables. They are handy when you want to pass information to a macro and don't want to add another parameter when you call the macro.

If you click the changing image up top, you'll be shown the source code that's behind the screen captures.

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